

Joseph Wallace

Computational Chemist | Molecular Modelling | Ligand Screening Workflows

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Professional Summary

Computational chemist with a background in molecular modelling, simulation, and structure-focused analysis across biomolecular and nanoscale systems. Currently building reproducible ligand-screening, docking, and rescoring workflows for drug-discovery-style prioritisation, with a strong focus on physically grounded modelling and benchmark-aware evaluation.

Core Skills

Molecular Modelling & Screening: Docking workflows, rescoring strategy, molecular interaction analysis, structure-based modelling, comparative ranking and benchmark evaluation.

Scientific Computing & Data Analysis: Python, NumPy, pandas, SciPy, matplotlib, reproducible analysis pipelines, workflow automation, data visualisation.

Simulation Methods: Molecular dynamics, enhanced sampling, umbrella sampling, metadynamics, Hamiltonian replica exchange, free-energy-oriented analysis.

Cheminformatics & Molecular Tools: RDKit, PyMOL, VMD, xTB.

HPC & Research Computing: Linux environments, HPC workflows, job submission, simulation pipeline management. Software: GROMACS, PLUMED.

Professional Experience

Postdoctoral Researcher

2022–Present

University of Padova / Italian Institute of Technology, Italy

- Lead computational modelling and analysis on nanoparticle–molecule systems; designed Python-based pipelines for automated processing, visualisation, and benchmark-style comparison across systems.
- Applied molecular simulation and enhanced-sampling methods to study structure, dynamics, and interaction behaviour in complex molecular systems, translating results into interpretable and decision-ready outputs.

PhD Research Attachment

2021–2022

Université Sorbonne Paris Nord, France

- Supported research continuity during laboratory relocation; contributed to computational research within an international academic environment.

PhD Research Attachment

2019–2021

A*STAR Institute of High Performance Computing, Singapore

- Conducted molecular simulation and structural analysis research in an HPC environment, with emphasis on scalable and reproducible workflows.

Education

PhD, Computational Biochemistry

2017–2022

University of Liverpool, UK

Thesis: How does constraining folded peptides within dense self-assembled monolayers on gold nanoparticles affect their structure? Combined molecular simulation and experimental data to study structure, dynamics, and interactions in peptide–nanoparticle systems.

BSc Biochemistry, First Class Honours

2014–2017

University of Liverpool, UK

Project: Solving the crystal structure of Cytochrome bc₁ for the development of antimalarials.

Selected Projects

Docking + Rescoring Benchmark for Ligand Ranking

- Built a reproducible workflow to test whether rescoring improves prioritisation over baseline docking on a curated protein–ligand benchmark.
- Focused on ranking analysis, benchmark design, and explicit failure-mode evaluation rather than inflated performance claims.
- Links:* [Project page](#) | [Repository](#)

Simulation Analysis Pipelines for Molecular Systems

- Designed Python-based workflows for processing, comparing, and visualising molecular simulation outputs across multiple systems.
- Emphasis on reproducibility, interpretable analysis, and workflow reuse.

Selected Publications

- Wallace, J., Riccardi, L., Mancin, F., & De Vivo, M. (2025). A Scoring Function for Monolayer-Protected Gold Nanoparticles Capable of Recognizing Small Organic Molecules in Solution. *Journal of Chemical Theory and Computation*, **21**(21), 11070–11078. DOI: 10.1021/acs.jctc.5c01278
- Franco-Ulloa, S., Cesari, A., Zanoni, G., Riccardi, L., Wallace, J., et al. (2025). Rational design of gold nanoparticle-based chemosensors for detection of the tumor marker 3-methoxytyramine. *Chemical Science*, **16**, 6282–6289. DOI: 10.1039/D4SC08758E
- Wallace, J. (2022). *How Does Constraining Folded Peptides within Dense Self-Assembled Monolayers on Gold Nanoparticles Affect Their Structure? An Experimental and Computational Investigation*. PhD thesis, University of Liverpool.